ISPRS Journal of Photogrammetry and Remote Sensing 195 (2023) 1–13

Contents lists available at ScienceDirect

ISPRS Journal of Photogrammetry and Remote Sensing

journal homepage: www.elsevier.com/locate/isprsjprs

Multi-spectral multi-image super-resolution of Sentinel-2 with radiometric consistency losses and its effect on building delineation

Muhammed T. Razzak [a,*], Gonzalo Mateo-García [b], Gurvan Lecuyer [c], Luis Gómez-Chova [b], Yarin Gal [a], Freddie Kalaitzis [a,*]

[a] Oxford Applied and Theoretical ML Group, University of Oxford, UK
[b] Image Processing Laboratory, University of Valencia, Spain
[c] Advanced Concepts Team, European Space Agency, The Netherlands

ARTICLE INFO

Dataset link: https://spaceml.org/repo/project/61c0a1b9ff8868000dfb79e1, https://github.com/ESA-PhiLab/RobustMFSRforEO, https://spacenet.ai/sn7-challenge/

Keywords:
Super-resolution
Multi-image super-resolution
Sentinel 2
Segmentation
Building detection

ABSTRACT

High resolution remote sensing imagery is used in a broad range of tasks, including detection and classification of objects. High-resolution imagery is however expensive to obtain, while lower resolution imagery is often freely available and can be used for a range of social good applications. To that end, we curate a multi-spectral multi-image dataset for super-resolution of satellite images. We use PlanetScope imagery from the SpaceNet-7 challenge as the high resolution reference and multiple Sentinel-2 revisits of the same location as the low-resolution imagery. We present the first results of applying multi-image super-resolution (MISR) to multi-spectral remote sensing imagery. We, additionally, introduce a radiometric-consistency module into the MISR model to preserve the high radiometric resolution and quality of the Sentinel-2 sensor. We show that MISR is superior to single-image super-resolution (SISR) and other baselines on a range of image fidelity metrics. Furthermore, we present the first assessment of the utility of multi-image super-resolution on a semantic and instance segmentation – common remote sensing tasks – showing that utilizing multiple images results in better performance in these downstream tasks, but MISR pre-processing is non-essential.

1. Introduction

Generative Deep Learning has sparked a new wave of Super-Resolution (SR) algorithms that enhance the spatial resolution of images with impressive aesthetic results (Wang et al., 2020). Although the perceptual quality of those images is high, it is well-known that some of these SR models introduce artifacts into the SR image that are not present in real images (Bhadra et al., 2020). In addition, most of these models do not enforce physically-based consistency between the super-resolved image and its low-resolution counterpart (Bahat and Michaeli, 2020). This limits the applicability of SR models to domains such as remote sensing where the safety and consistency are critical, e.g. for scientific instrumentation and decision making.

Super-resolution models can be divided in *single-image* super-resolution (SISR) and *multi-image* super-resolution (MISR) (aka *multi-frame* or *multi-temporal* super-resolution). The former uses as input only one low-resolution image while the later takes several low-resolution images from the same scene. MISR seeks to further constrain the ill-posed problem of SR by conditioning on several low-res input images (aka revisits). Therefore, it is expected of MISR to produce better SR images, to be more robust, and to produce fewer artifacts than SISR. In addition, MISR can be naturally applied in Earth observation since satellites often have frequent revisits of an area of interest. These multiple revisits can be fused with MISR to produce a super-resolved image. Despite its clear applicability, MISR has been scarcely applied in Remote Sensing and, as of yet, there are no studies that quantitatively compare MISR and SISR. Thus far, for remote sensing, MISR has been demonstrated only on RED and NIR bands of PROBA-V –a tiny fraction of the Sentinel-2 operation spectrum (Deudon et al., 2020a). In addition, the practical utility of these super-resolved images for downstream tasks has been largely unexplored in real-world applications (Shermeyer and Van Etten, 2019).

In this work, we apply MISR to the multi-spectral multi-temporal satellite imagery from the European Space Agency's Copernicus Sentinel-2 (S2) archive, and study the downstream utility of the super-resolved images. In particular, we train super-resolution models (both SISR and MISR) on the 10 m RGB bands of Sentinel-2 images, using as reference co-registered 4.77 m RGB PlanetScope images, acquired within the same two-month period. This setting differs from the vast

* Corresponding authors.
E-mail addresses: muhammed.razzak@cs.ox.ac.uk (M.T. Razzak), gonzalo.mateo-garcia@uv.es (G. Mateo-García), gurvan.lecuyer@esa.int (G. Lecuyer), luis.gomez-chova@uv.es (L. Gómez-Chova), yarin@cs.ox.ac.uk (Y. Gal), freddie.kalaitzis@cs.ox.ac.uk (F. Kalaitzis).

https://doi.org/10.1016/j.isprsjprs.2022.10.019
Received 4 April 2022; Received in revised form 27 October 2022; Accepted 31 October 2022
Available online 12 November 2022
0924-2716/© 2022 The Author(s). Published by Elsevier B.V. on behalf of International Society for Photogrammetry and Remote Sensing, Inc. (ISPRS). This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).



Fig. 1. Different co-registered acquisitions from PlanetScope and Sentinel-2 from the SpaceNet 7 dataset. First row: PlanetScope RGB. Second and third rows: Sentinel-2 RGB revisits. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

majority of previous remote sensing applications of SR, where low-res images are obtained by artificially downsampling the high-res counterpart (Shermeyer and Van Etten, 2019). The main benefit of our setting is that the trained model can be applied on new S2 RGB images to enhance their nominal resolution to 4.77 m, i.e. it provides out-of-sample SR results without requiring simultaneous and co-registered VHR images. We compare this model with SISR models on an independent test set showing better qualitative and quantitative performance in terms of PSNR (Peak Signal to Noise Ratio) and SSIM (Structural Similarity Index Measure), see Section 6.1. In addition, we demonstrate that this resolution enhancement further produces significant gains in tasks such as building delineation (Section 4).

One of the drawbacks of training super-resolution models when the low-res and high-res images come from different sensors, is the difference in the spectral characteristics of the two sensing instruments and in the calibration of their output images. This poses the additional challenge of disentangling the SR task from the cross-instrument calibration task. Fig. 1 shows several co-located Sentinel-2 and PlanetScope images: for each PlanetScope image, we show four S2 revisits within the same two-month period. We can see that there are differences in the colors of these images, in particular, Sentinel-2 images are brighter and with higher contrast and colors seem better defined. This is because Sentinel-2 images benefit from a higher radiometric and spectral resolution compared to PlanetScope, and both products undergo different atmospheric correction procedures to recover surface reflectance. In order to produce super-resolved images while preserving the spectra of the low-res Sentinel-2 imagery, we propose a novel radiometric consistency module to the super-resolution model, that helps to maintain the radiometric consistency of a super-resolved image with its low-res counterpart.

The contributions of our work are summarized as follows:

1. We curate a new multi-temporal dataset from many revisits of Sentinel-2 imagery co-located with PlanetScope imagery, originally sourced from the SpaceNet-7 competition (Van Etten et al., 2021), which includes a geodiverse set of scenes from around the globe.
2. We demonstrate, for the first time, the multi-image super-resolution of RGB satellite imagery.
3. We show that MISR is better at super-resolving Sentinel-2 RGB images compared to SISR, both quantitatively in terms of the image fidelity metrics (PSNR and SSIM) and qualitatively.
4. We demonstrate the downstream utility of super resolution, through the task of building semantic segmentation and instance segmentation.
5. Finally, we propose a radiometric consistency module which can be added to any vanilla super-resolution model. We show that this module helps to maintain the radiometric consistency of Sentinel 2 while enhancing its spatial resolution, and we show several instances of good and bad consistencies.

The new multi-temporal dataset of Sentinel-2 and PlanetScope imagery, and the training, validation and test sets used in the experiments are presented in Section 2. Section 3 describes the single-image and multi-image super-resolution methods analyzed in this work. Section 4 presents the building detection downstream task used to demonstrate the utility of super-resolved images. Section 5 describes the modified MISR model including the proposed radiometric consistency module. Finally, experimental results and conclusions of the work are discussed in Sections 6 and 7, respectively.

2. Materials and datasets

In order to learn a supervised super-resolution model to improve the spatial resolution of S2, we need higher-resolution images to be used as a reference. Since VHR (Very High Resolution) images (less than 10 m) are not freely available, we restricted our search to pre-released publicly-available datasets of high-resolution images. Among those, we chose the recently launched Multi-temporal urban development SpaceNet dataset of PlanetScope images (also known as SpaceNet-7, see Section 2.1) (Van Etten et al., 2021). Given this dataset, we acquired co-located time series of Sentinel-2 images for each PlanetScope acquisition (Section 2.2). Section 2.3 has a brief analysis of the S2-Planet dataset as well as details about the different train–test splits that we used for the results.

2.1. PlanetScope SpaceNet-7 dataset

SpaceNet-7 (Van Etten et al., 2021) has monthly time series of PlanetScope images over a two-year time span period for approximately 100 different areas of interest (AOI) all over the world (see Fig. 2). Images are provided at 4.77 m nominal[1] resolution with only three

[1] The resolution reported in the GeoTIFF metadata.

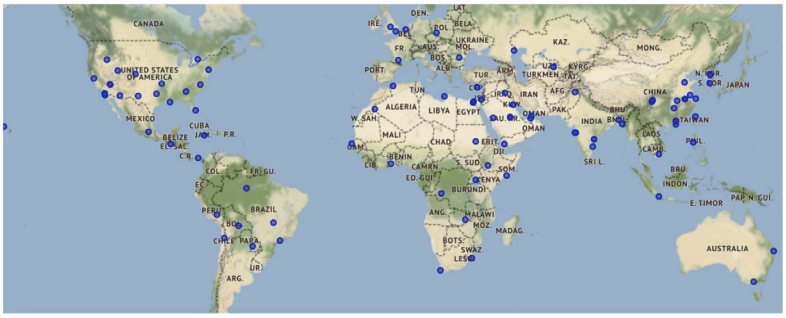


Fig. 2. Location of SpaceNet-7 image time series.
Source: Figure taken from Van Etten et al. (2021).

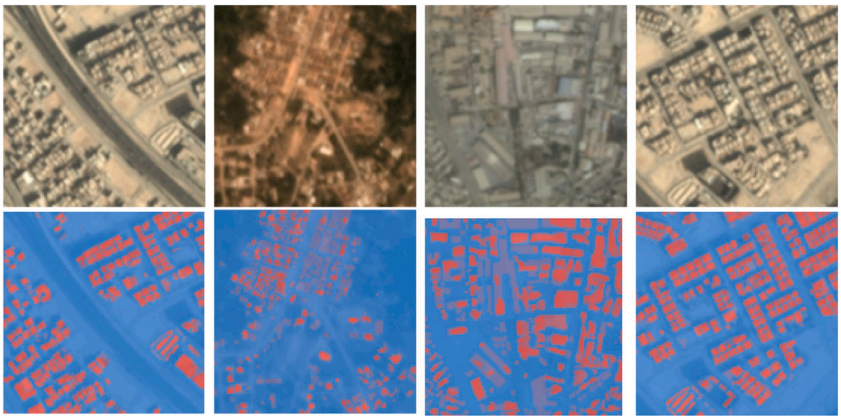


Fig. 3. PlanetScope acquisitions (first row) and their corresponding building polygon masks (second row) from the SpaceNet-7 dataset.

spectral channels (RGB) and size around 1000 × 1000 pixels. These images were sourced from Planet's global monthly basemaps which are mosaics of best scenes selected according to some quality metrics from PlanetScope Dove constellation as covered in Van Etten et al. (2021). The imagery is encoded in 24 bits (per pixel) and there is no information on the calibration or atmospheric correction of those images. Each of those images have manually derived polygon labels of building footprints. The challenge accompanying the release of this dataset consisted of delineating those buildings and identifying when new buildings were constructed on those areas. Fig. 3 shows some PlanetScope acquisitions with their corresponding building annotations over different locations. We can see that delineating the buildings on some of these scenes is quite challenging even at the resolution of PlanetScope. In this study, we restricted to images from December 2019 and January 2020 from the training set (building footprints in the test set have not been released). In total there are 45 different PlanetScope scenes for each month.

2.2. Sentinel-2 acquisitions

The Sentinel-2 mission consists of two twin satellites carrying the same multi-spectral optical sensor which acquires images on 13 different bands of the electromagnetic spectrum, from the visible to the short-wave infrared. The nominal spatial resolution of those images is different for each set of bands: 4 bands (visible and near infra-red) have 10 m resolution; 6 bands in the very near infrared and short-wave infrared have 20 m resolution, and the remaining 3 bands, which are used mainly for atmospheric correction, have a spatial resolution of 60 m. Level 2A Sentinel-2 products, often referred as analysis ready data, consist of atmospherically corrected orthorectified 12-band images with bottom-of-atmosphere (BOA) calibrated reflectance.[2] Providing good BOA calibrated images is one of the main goals of the Sentinel-2 mission, since an accurate radiometric and spectral calibration has large impacts on ocean (see e.g. Ruescas et al., 2018; Sauzède et al., 2020) and vegetation bio-physical products (e.g. Wolanin et al., 2019; Svendsen et al., 2018). In Section 5, we propose a SR model that seeks to maintain the spectral calibration of Sentinel-2 in the SR images.

Sentinel-2 images were downloaded from the European Space Agency (ESA) Open Access Hub. In order to obtain co-aligned time series of Sentinel-2 and PlanetScope images, we developed a custom pipeline which consists of the following steps:

1. Download all Sentinel-2 level 2A products overlapping with each of the 45 PlanetScope scenes over December 2019 and January 2020.
2. Crop all Sentinel-2 images to the PlanetScope scene bounds.
3. Reproject all bands of S2 to the coordinate reference system of PlanetScope products at 10 m spatial resolution.
4. When more than one S2 product was found for the same date and scene, we mosaiced those images.

Fig. 1 shows different Planet (top) and Sentinel-2 (bottom) acquisitions. As we can see, some of the Sentinel-2 images present clouds. Sentinel-2 products have additional quality assessment bands that include cloud probabilities for each pixel. In this work, we used the SLC band to assess the presence of clouds; in particular, we used the value 9 which encodes 'clouds high probability'. This cloud indicator is used to inform the fusion model when merging different Sentinel-2 revisits.

[2] The B9 band used for cirrus detection is not included in Level 2A products.

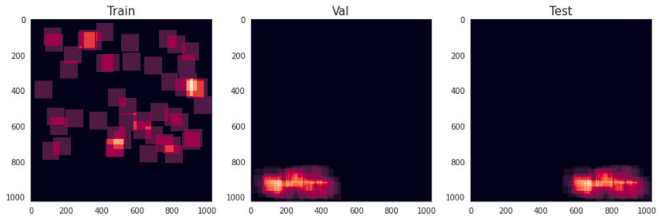


Fig. 4. We utilized a within-scene split, which allocates the top 80% of a scene as a source of training patches; the bottom-left and right 10% for validation and testing patches respectively.

2.3. Dataset analysis and training splits

The images come from 45 locations, with a geodiverse set of features including vegetation, bare soils (flats, hills, ridges), desert, urban, and agriculture infrastructure (see Appendix Table A.1). The number of revisits between December 2019 and January 2020 range from 5 to 13 and the percentage of usable revisits (< 50% cloud coverage) ranges from 23% to 100%.

2.3.1. Training, validation and test sets splits

A well-thought split of the data into training and testing is critical to demonstrate the capacity of machine learning models to generalize. In remote sensing scenarios, extra-care must be taken to avoid train–test leakage due to spatial correlation. For instance, Ploton et al. (2020) recently showed that lack of consideration to spatial correlation lead to over-inflated results of ML models that monitored forest biomass. In this work, we test the models using two different training and testing splits. In the first approach, we split each scene in patches avoiding spatial overlap between patches in the different subsets; with this approach, we seek to explore the performance of the models in ideal conditions when training and testing patches come from similar distributions. Fig. 4 shows the dataset partition for one scene following this approach. In the second approach, models are tested in images from the same location but different time periods (one month before). This split seeks to explore the capacity of the models to generalize to different time acquisitions.

3. Multi-spectral multi-image super-resolution

In the context of remote sensing, our work is the first to tackle the MISR problem in multiband images (RGB bands). Although the ultimate goal is multi-spectral MISR in all 13 bands of Sentinel-2, such a task is predicated on the coverage of the same bands by the high-resolution instrument, at least in the proposed supervised learning setting. Unfortunately, most VHR (very high resolution) products are limited to Red, Green, Blue, and NIR (near-infrared) bands.

Back in traditional on-the-ground imaging, MISR has seen several successful applications on color photos. Most notably, Wronski et al. (2019) achieved real-time on-board multi-image super-resolution in handheld Google Pixel cameras, by leveraging the user's hand motion jitter during a burst-frame shot. With intimate knowledge of the handheld camera's specifications (physical model), and no feature-learning involved, the authors could anticipate the amount of aliasing and phase shifting that images undergo after optical zoom-out and motion-jitter, respectively. The fusion of burst-frames is ultimately a convex optimization problem (ADMM, see Boyd et al. (2011)). More recently, a deep learning approach to bust-frame MISR was proposed in Bhat et al. (2021), which is agnostic to camera specifications.

3.1. MISR in earth observation

In handheld camera imaging, the problem of MISR is always learned with high-res reference images that comes from the same sensor. However, in Earth Observation, the fact that any one instrument orbits at a fixed altitude, rules out any possibility of obtaining simultaneous low-res/high-res images pairs by the same sensor. Hence, this necessitates the use of a different higher resolution instrument, if MISR is to be learned in a supervised way. So unless the low-res and high-res pairs are acquired with different lens on-board the same satellite (e.g. Proba-V, as a unique example at that, see Märtens et al. (2019)), then any ML-based MISR model must also learn to calibrate its output to the spectra and noise of the high-res instrument. In addition, for remote sensing, the multiple images are temporally spaced over days and weeks rather than over a few seconds with handheld imagery.

Recently, Molini et al. (2020) and Deudon et al. (2020a) tackled the MISR problem in Earth Observation, in single-band imagery, and on the rather controlled use-case of Proba-V that was the topic of ESA's MISR competition ending Q2-2019. In particular, Deudon et al. (2020a) was the first approach to tackle the different problems in MISR (input co-registration, fusion, and registration-at-the-loss) in an end-to-end manner, and with a small memory footprint (due to its reused fusion operator in the low-res domain). Since then, several deep learning approaches with refined architectures have repeatedly beaten the state-of-the-art in the Proba-V "post-mortem" leaderboard; most notably (Salvetti et al., 2020). Closer to our application area, Kawulok et al. (2021) apply a non end-to-end MISR method to Sentinel-2, containing 1 band, but primarily focus on artificially degraded imagery.

3.1.1. HighRes-net

This work modifies the HighRes-net architecture of Deudon et al. (2020a) to the S-2 and PlanetScope RGB images described in Sections 2.1 and 2.2. For a complete description of HighRes-net, we refer the reader to the original paper (Deudon et al., 2020a). Table B.1 is a modular breakdown of the HighRes-net architecture. The hallmark feature of HighRes-net is its shared fusion operator that can be pairwise-applied recursively on an arbitrary set of encodings of low-res revisits. This technique can be easily parallelized on a GPU, through careful use of the `torch.Tensor.view` operator, that treats the different encoding-pairs as instances of a mini-batch.

The next step, ShiftNet,[3] is not intrinsic to HighRes-net but an ancillary learned image registration operator to account for the inevitable misalignments that are explained by shifts.

3.2. Single-image super-resolution

Single-image super-resolution (SISR) has progressed significantly with the advent of deep learning, and has progressed even further with newer developments. Initial approaches to super-resolution with deep architectures were CNN-based. First, SRCNN (Dong et al., 2015) followed by FSRCNN (Dong et al., 2016), which improved the speed of SRCNN along with a minor gain in performance. These approaches used some form of upsampling (transposed convolutions or bicubic upsampling) interspersed with convolutional layers to achieve the improved super-resolved imagery. The objective function used to train these networks is the mean square error between the high-resolution ground truth image and the super-resolved output image.

The work of Ledig et al. (2017) provided a significant step forward in terms of photo-realistic and perceptually pleasing super-resolution. They introduced a far better CNN-based super-resolution network called Super-Resolution Residual Networks (SRResNet). SRResNet was deeper than prior works and included residual blocks. Additionally,

[3] A reduction of HomographyNet from 8 parameters for homographies to 2 for translations, see DeTone et al. (2016).

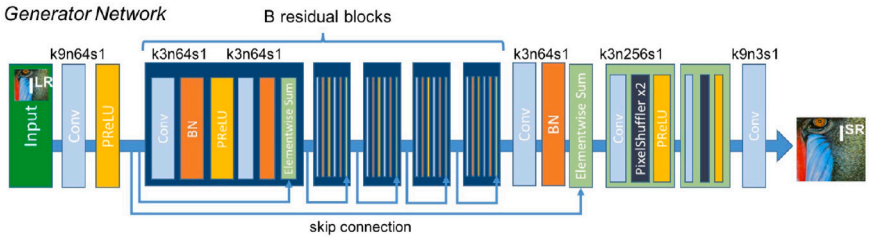


Fig. 5. SRResNet model architecture as illustrated in Ledig et al. (2017).

instead of transposed convolutions or bicubic upsampling SRResNet made use of pixel shuffling to upsample the imagery. However, the major contribution of Ledig et al. (2017) was the introduction of a generative adversarial network for super-resolution (SRGAN) that achieved perceptually pleasing results. SRGAN uses SRResNet as the generator and a simple CNN for the discriminator. In Ledig et al. (2017), they note that while SRResNet, trained with an MSE, achieves superior performance in terms of PSNR and SSIM to all other methods including SRGAN, SRGAN is able to achieve more perceptually pleasing results (capturing high-frequency content) as evaluated by Mean Opinion Score (MOS) of a panel of human evaluators.

More recently, there has been further work on improving super-resolution through GAN training (Wang et al., 2018; Anwar et al., 2020; Wang et al., 2020), and flow-based approaches (Lugmayr et al., 2020). These developments have been also adapted to the field of Earth observation in different applications: A GAN-based edge-enhancement network was proposed in Jiang et al. (2019) for super-resolution reconstruction. A multiband GAN-based SR method was presented in Li et al. (2020) to avoid the spectral distortion and exploit the correlation of spectral bands. Latte and Lejeune (2020) do radiometric normalization between PlanetScope and Sentinel-2 imagery with single images, while super-resolving Sentinel-2 imagery. Finally, a visual saliency GAN presented in Zhang et al. (2020) improved the perceptual quality by avoiding undesired artificial textures.

3.2.1. SRResNet

While we examined many architectures as a SISR baseline to compare HighRes-net against, we decided to restrict the comparison to SRResNet for a couple of reasons:

1. It is amongst the best performing SISR methods in terms of PSNR and SSIM. These are the primary metrics we will be evaluating our results on. We are interested in accuracy over perceptually pleasing results, and also had no possibility to evaluate with a panel of human evaluators to obtain an MOS.
2. SRResNet, in terms of representational capacity, is similar to that of HighRes-net. Both networks rely upon residual blocks to encode the images. SRResNet does however make use of pixel shuffling instead of transpose convolution. Pixel shuffling has been shown to do better upsampling in Ledig et al. (2017).

In addition, when benchmarking MISR against SISR, it is important to define the *Single-Image Selection* strategy used in SISR. In particular, given the high Sentinel-2 revisit, one has to choose a revisit on which to do super-resolution using an SISR model. One could choose a random revisit, however, in order to have an stronger baseline, we choose the best revisit indicated by the lack of clouds.

3.3. Super resolution metrics

For super-resolution, the primary quantitative *metrics of performance* are the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM) (Wang et al., 2004) between the super-resolution image and the reference high-resolution image. Both measure the fidelity of the compared images, but with SSIM more focused on the structures contained in the images. Mathematically, PSNR is computed as follows:

$$PSNR = 10 \log \frac{I_{max}}{\mathrm{MSE}(HR, SR)} \tag{1}$$

where I_{max} is the maximum pixel value of the images (e.g. for 8-bit images this is 255), MSE is the mean square error between the high-resolution image (HR) and super-resolved image (SR). For the formula of the SSIM the reader is referred to the original work of Wang et al. (2004).

We evaluated three *objective functions* during training: Mean Square Error (MSE), Mean Absolute Error (MAE) and Structural Similarity Index Measure (SSIM). While MSE was able to obtain superior performance in terms of PSNR the imagery was far less sharp. Using SSIM as the objective function resulted in excellent PSNR performance, superior SSIM performance and imagery that was sharper and perceptually pleasing.

SRResNet's model architecture is illustrated in Fig. 5, the rest of the hyper-parameters used to train the network are described in the Appendix C.1.

4. Utility of MISR for building delineation

The interplay between super-resolution techniques and a range of downstream task performance (object detection, instance segmentation, semantic segmentation) remains largely unexplored, particularly in the context of satellite imagery. In Shermeyer and Van Etten (2019), authors performed single image super-resolution using synthetically downscaled VHR WorldView-3 images. Afterwards, they compared the performance on an object detection task using images at different resolutions (real, downsampled and super-resolved images). They showed an increase in performance when using super-resolved images instead of artificially downsampled ones. This was far from a real-world scenario given the synthetic data used and moreover it was only assessed on a single task and dataset.

Detecting objects such as buildings from Sentinel-2 or even PlanetScope imagery, which have resolutions of 10 m and 4 m respectively in the visible spectrum, is a difficult task as these objects often cover a small amount of area in terms of pixels. In the case of buildings in urban areas, they are also densely packed making the task of delineating between buildings even more difficult. Increasing the spatial resolution thus lead to better detection and delineation of these objects.

In this study, we apply the previously trained super-resolution models to the downstream task of the SpaceNet dataset: semantic and instance segmentation of buildings. We compare the models performance on the native (ground-truth) imagery and super-resolved imagery at the same ground sampling distance, and the low-resolution imagery with bicubic upsampling.

Ours is the first study to assess the utility of MISR, and the first study to assess the utility of SISR or MISR on real-world data, as opposed to synthetically downsampled imagery.

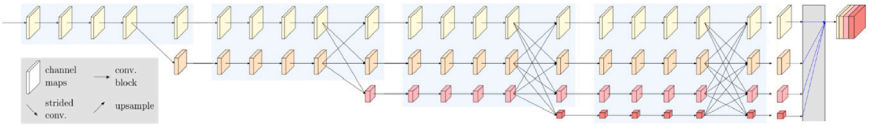


Fig. 6. HRNet Architecture: high-resolution representations for Semantic Segmentation (Yuan et al., 2020).

4.1. Downstream models and processing pipeline

The state-of-the-art in building footprint segmentation has converged to a common algorithm: identify instances and extract polygons. The *identify instances* step has been approached as instance segmentation task, in which instances are directly obtained from an instance segmentation network, or more commonly as a semantic segmentation task, in which instances are obtained using the connected components of the mask predicted by a semantic segmentation network. In both approaches, the *extract polygons* step is then carried out for each instance by a simple vectorization routine.

We follow the second avenue, semantic segmentation, as this was the approach used among all four winners of the SpaceNet-7 challenge.

4.1.1. Input imagery

As input imagery for the building delineation task we use the MISR or SISR models outputs, trained as previously described in Section 3.1, in addition to a number of baselines:

1. Bicubic upsampling: The clearest and less cloudy revisit of Sentinel-2 imagery was bicubicly upsampled.
2. SISR: The clearest revisit of low-resolution Sentinel-2 imagery was fed through a trained SRResNet.
3. Multi-Image upsampling: The clearest four (4) revisits of low-resolution Sentinel-2 imagery were bicubicly upsampled and stacked.
4. MISR: All revisits of the low-resolution Sentinel-2 imagery were fed through HighRes-net.
5. PlanetScope: The high-resolution PlanetScope imagery.

4.1.2. Semantic segmentation

Semantic segmentation is the task of assigning each pixel a semantically meaningful label. In this setting, it is specifically the task of assigning whether a pixel belongs to a building or not. In this work, the aim of this evaluation is to compare the same semantic segmentation model using the different inputs referred before to assess the practical utility of the super-resolved images. For this purpose, we chose to use HRNet (Yuan et al., 2020) as the semantic segmentation model which is trained using all the input configurations of sub Section 4.1.1. We chose HRNet because it is the network used by the winning solution in the SpaceNet-7 challenge, and is amongst the state-of-the-art networks as evaluated on variety of benchmarks (Wang et al., 2019). The architecture for the network is illustrated in Fig. 6; we list in Appendix C.2 the hyper-parameters that we used for training the models (note that we did not perform a comprehensive hyper-parameter sweep).

Semantic segmentation metrics. The primary metric of semantic segmentation is Intersection-over-Union (IoU), which is evaluated for every class. The mean IoU over all the classes is the final reported metric. IoU is a measure of overlap between the predicted per-pixel segmentation and the ground truth divided by the area of union between this segmentation and the ground truth. IoU can be calculated as follows using either set or confusion matrix notation:

$$IoU = \frac{A \cap B}{A \cup B} = \frac{TP}{TP + FP + FN} \tag{2}$$

where A is the predicted mask of the buildings and B is the ground-truth mask.

4.1.3. Instance segmentation

In the instance segmentation task, we aim to delineate each building instance. We do this through a simple process of vectorization that derives polygons using the connected components of the output of the semantic segmentation model. This is the standard method of polygon extraction and has been used within the SpaceNet challenge along with industrial uses by the likes of Microsoft for their maps.

Instance segmentation metrics. The primary metrics for instance segmentation are precision, recall, and F1 score. These metrics are calculated on the produced vector instances (polygons) rather than in the pixels. For a positive detection (true positive), the IoU between the prediction and ground-truth polygons must be greater than a threshold. For this particular dataset, the IoU threshold in the competition was set at 0.25. Consequently, we used that threshold in our evaluation too.

5. Color matching HighRes-net for sentinel-2 MISR

The Sentinel-2 level 2A products used as inputs consist of orthorectified images of BOA reflectance (see Section 2.2). In contrast to the building delineation task, which mainly relies on the spatial and morphological information, a large number of land and ocean S2 applications rely on well-calibrated reflectance values to provide meaningful outputs (see for instance Ruescas et al., 2018; Wolanin et al., 2019; Svendsen et al., 2018). However, the proposed MISR training scheme it is optimized to minimize disagreement between the PlanetScope and the super-resolved output image. Since PlanetScope images have a different color calibration than S2 (PlanetScope images from the SpaceNet-7 dataset are in fact not calibrated, see Section 2.1), we experienced that the SR images *look like* PlanetScope images in terms of color (see Fig. 1 to appreciate the difference in color between S2 and PlanetScope). Hence, by using the proposed MISR model, somehow we loose the radiometric calibration and atmospheric correction of S2 L2A images. In this section, we propose a modification of the MISR model and training procedure to produce spectrally-consistent S2 super-resolved images.

Fig. 7 shows a diagram of the modified forward path with the consistency loss. On the top of the image, we have HighRes-net as proposed in Deudon et al. (2020b): this network fuses a set of low-res revisits using the encoder (recursive fusion) and afterwards it applies the decoder to the fused feature-maps (upsample). However this output (SR in the figure), instead of being compared with the high-resolution image to compute the loss, it is further processed in two losses that are averaged and minimized together. For the first loss, that we called (color/spectral) *consistency loss*, we downsample the SR image back to the low-res size and compare it with the original S2 reference frame. For the second loss, called *super-resolution loss*, we apply a pixel-by-pixel fully connected neural network (implemented as 2 layers of 1×1 convolutions) to produce the *PlanetScope like* output. We refer to this network as *color matching network*. By using this additional network we seek to disentangle MISR (controlled by HighRes-net) from spectral matching (learn by the color matching network). We were inspired by the works of Tasar et al. (2020) and Mateo-García et al. (2020) that use similar ideas in the context of domain adaptation.

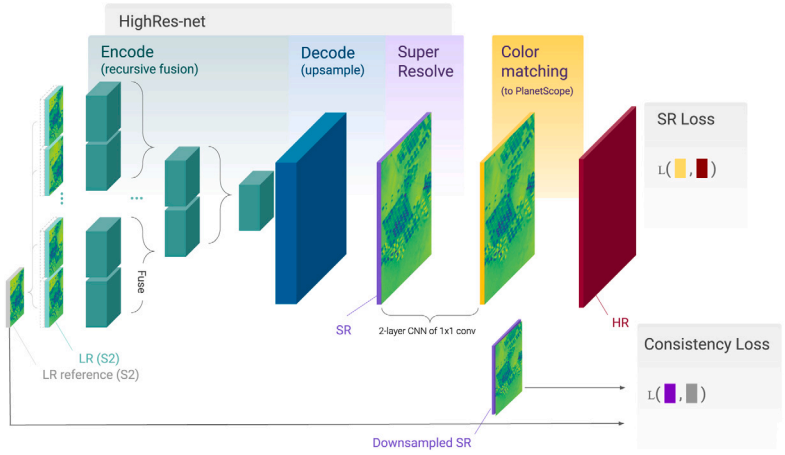


Fig. 7. HighRes-net with a consistency loss between a downscaled version of the SR output and the low-res reference frame.

Table 1
Average PSNR and SSIM scores (higher scores are better).

Subset	Bicubic		SRResNet		HighRes-net	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Train	17.85	0.612	27.90	0.827	**30.16**	**0.88**
Validation	18.11	0.70	26.06	0.78	**28.52**	**0.84**
Test	18.54	0.70	26.83	0.80	**29.40**	**0.85**

5.1. Implementation details

For the consistency loss, we used the MSE between the downsampled image and the low-res reference, the downsampling step uses adaptive average pooling. The color matching network consists of a 1×1 convolutional layer with 64 output channels, a ReLU activation and a 1×1 conv layer back to 3 output channels. This output is shifted using the ShiftNet network, explained in Section 3.1.1, and compared with the PlanetScope high-res image using an SSIM loss. Finally, the consistency loss and the super-resolution loss are averaged together using a convex combination of 0.9 and 0.1, respectively.

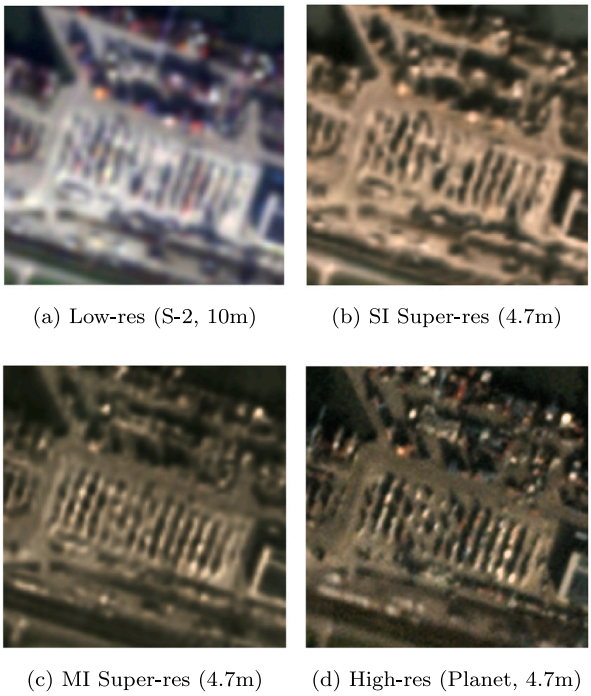
6. Experimental results

In this section, we present the results of the Multi-Image Super-Resolution performance and image reconstruction quality; the building delineation results of the selected downstream task; and the color consistency results of the super-resolved Sentinel-2 images.

6.1. Multi-image super-resolution results

This section focuses on performance of the models in terms of Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) (cf. Section 3.3). These are the most commonly used metrics for measuring super-resolution performance and image reconstruction quality.

Table 1 shows the super-resolution performance of our evaluated models with the split described in Section 2.3.1. *Bicubic* method indicates the scores if LR images are just upscaled using bicubic interpolation to match the HR image size. Bicubic interpolation and SRResNet (SISR) are computed using the best S2 revisit in terms of cloud coverage. Here we see that the performance of the MISR method (HighRes-net) is significantly higher than SISR (SRResNet) and Bicubic interpolation. Additionally, we see that scores calculated in training and testing splits do not differ much in the trained models (both in SSResNet and HighRes-net) which shows low overfitting.



(a) Low-res (S-2, 10m) (b) SI Super-res (4.7m)

(c) MI Super-res (4.7m) (d) High-res (Planet, 4.7m)

Fig. 8. Patch from a validation-set scene. The super-resolved output of HighRes-net 8(c) better delineates the buildings in urban scenes, hence enabling downstream tasks like building segmentation, with improved accuracy compared to prediction on a single S2 image 8(b) or the bilinearly upsampled Sentinel-2 image 8(a).

Table 2
Average PSNR and SSIM scores for testing on a different time period.

Subset	Bicubic		SRResNet		HighRes-net	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Train	17.27	0.63	22.66	0.74	**23.28**	**0.76**
Validation	18.14	0.69	22.49	**0.73**	**22.56**	**0.73**
Test	16.99	0.66	22.12	0.73	**23.10**	**0.77**

In addition, we tested the same model applied to acquisitions in a different time period but over the same regions. The results are shown in Table 2. We notice a drop in performance across all methods, but the super-resolution models still exceed the performance of the bi-cubic upsampling method by a significant margin. Additionally, we still see that MISR outperforms SISR in all subsets.

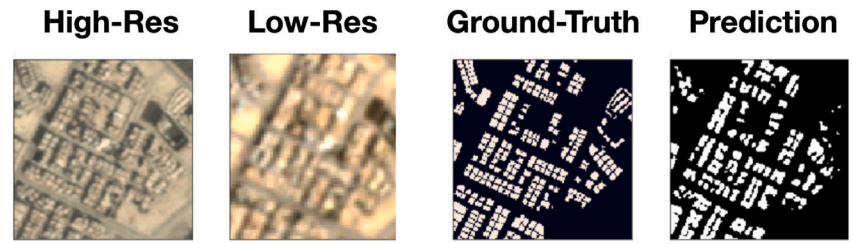


Fig. 9. A sample result of semantic segmentation on SpaceNet 7 dataset using MISR imagery from Sentinel-2.

Table 3
Performance of semantic segmentation models trained with different input imagery. On-par top accuracy shown in bold.

Model	IoU (val)	IoU (test)
S-2 (best revisit): Bicubic	0.60	0.60
S-2 (best revisit): SISR	0.58	0.56
S-2 (best 4 revisits): bicubic + concat	**0.62**	**0.62**
S-2: MISR (all revisits)	**0.61**	**0.60**
PlanetScope	0.66	0.69

Table 4
Performance of instance segmentation algorithm with different input imagery on the test set.

Model (IoU Threshold: 0.25)	Precision	Recall	F1
S-2 (best revisit): Bicubic	0.24	0.56	0.33
S-2 (best revisit): SISR	0.21	0.56	0.30
S-2 (best 4 revisits): bicubic + concat	**0.28**	**0.57**	**0.37**
S-2: MISR (all revisits)	**0.26**	**0.57**	**0.36**
PlanetScope	0.41	0.68	0.51

6.2. Building detection results

In this subsection, we present the results of the downstream tasks. Due to the manner in which the tasks are completed, they are very much linked.

Semantic segmentation. In the semantic segmentation results, we see that, as expected, the semantic segmentation model (HRNet) using MISR as input exceeds the performance of the bicubic upsampling and SISR. However, the model using multiple revisits (with bicubic upsampling) is able to match the performance of the MISR model. This result shows the utility of using multiple revisits, while at the same time showing that MISR (or at least our particular implementation) does not necessarily result the optimal representation for performance on the downstream task. Optimizing an HRNet model by directly using the best 4 revisits results in equal or better performance.

Additionally, we see that SISR underperforms against bicubic upsampling. We suspect this is likely due to SISR introducing artifacts, which results in the lower performance.

Finally, as expected, the PlanetScope imagery (used as ground-truth) resulted in a model with far superior performance than any of the other methods (see Table 3).

An example of the resulting semantic segmentation output given the inputs is shown in Fig. 9.

Instance segmentation. The results on the instance segmentation task largely mimic the results seen in the semantic segmentation. We can, however, see that performance margin grows between the model using the PlanetScope imagery and those using Sentinel-2, indicating the performance of the fine point accuracy required for correct delineation for instance segmentation on this particularly dataset.

We also see that precision is generally lower with the super-resolution imagery than with their equivalent bicubicly upsampled baselines. This indicates the super-resolution models are producing predictions that are less pure. One important note to make is that we used all revisits with the MISR model (including cloudy revisits), while the bicubicly upsampled model only used the best four revisits. While the MISR model was shown to be relatively robust to this, an avenue for further investigation should include using simply the best revisits for the MISR model (see Table 4).

Overall, we saw that MISR was shown to be useful for the resulting downstream tasks of semantic and instance segmentation. These results suggest that the obtained MISR images correctly encode the information contained in the Sentinel-2 time series.

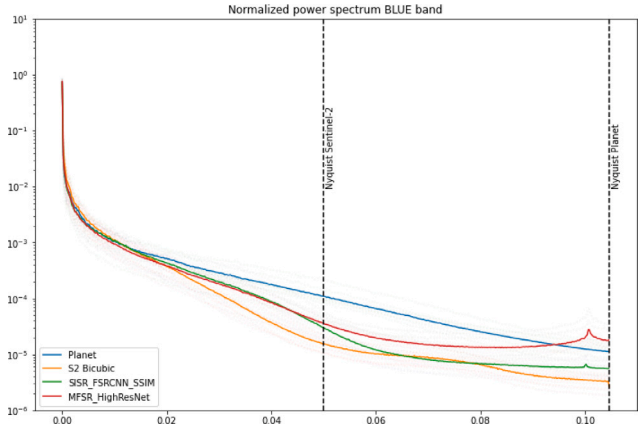


Fig. 10. Averaged power spectrum of PlanetScope, Sentinel-2 interpolated and Sentinel-2 super-resolved images over the train and validation dataset. We can see that super-resolved images using MISR and SISR (green and red) have more high-frequency components than Sentinel-2 (orange). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6.3. Color consistency results

In this subsection, we present the results of the proposed spectral consistency loss in terms of spatial frequencies, reflectance values, and perceptual quality.

Averaged spatial fourier power spectrum. Fig. 10 shows the spatial Fourier power spectrum averaged over the images in the test and validation datasets. In particular, we show the spectrum of PlanetScope images, Sentinel-2 images upscaled with bicubic interpolation and the super-resolved images of the MISR and SISR methods. We can see that super-resolved images (red and green) exhibit a larger amount of higher-frequency components than Sentinel-2 images (orange). This shows that the SR models are adding higher-frequency content to the output. On the other hand, still, the amount of high-frequency information is lower than in PlanetScope (blue).

Radiometric consistency. In order to evaluate the radiometric consistency loss term, Fig. 11 shows the histograms over the test and val datasets of each of the RGB bands. In these histograms we can see that the color distribution of PlanetScope (first row) is very different of the reflectance of Sentinel-2 (last row). In the middle, we show

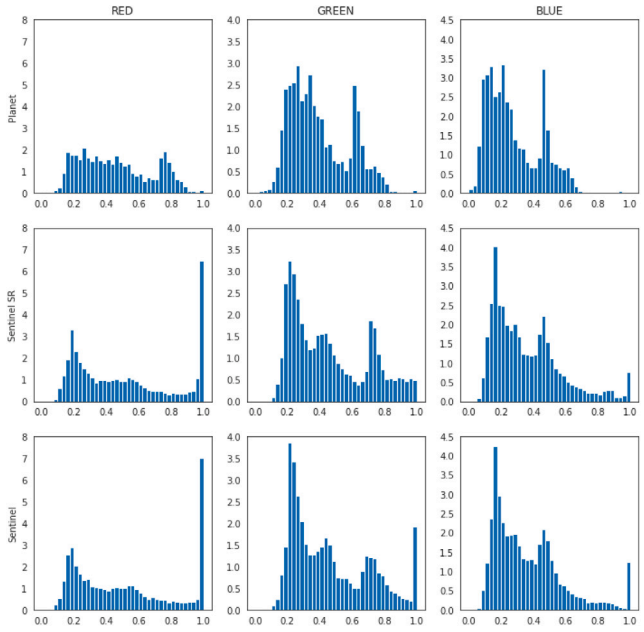


Fig. 11. Normalized histograms of the image values over the test and val dataset for each of the RGB bands (columns). First row shows the values of PlanetScope images, second row shows the values of the super-resolved Sentinel-2 images using the consistency loss. Third row shows the values of the Sentinel-2 images. We see that, using the color consistency loss, the color distribution of Sentinel-2 is maintained in the super-resolved images. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the histograms of the Sentinel-2 super-resolved images using the MISR method with consistency loss (Section 5). We can see that the RGB distribution of the SR images matches the distribution of Sentinel-2.

6.4. Static versus dynamic scenes

In this subsection, we present the result of a study conducted to measure the impact of the dataset on the performance of the MISR method. The super-resolution task relies on using multiple images acquired at different dates as input. Therefore, the revisits of the same scene can show significant changes caused by various factors such as vegetation, human activities, or weather events.

We separated the dataset into two subsets for this experimentation: a static set containing scenes with revisits showing no or very few changes and a dynamic set containing scenes with revisits visible difference. For each scene, we computed the average pixel variance over the revisits. We used the median average pixel variance as a threshold to discriminate between static and dynamic scenes. The final static scene contained twenty-three scenes against twenty-two for the dynamic scene. Most of the changes observed in the dynamic scenes were due to weather effects such as cloud shadow, or changes in the illumination conditions. We trained five MISR models on the static subset and five on the dynamic subset for this experimentation. Each model was then tested on both subsets individually, following the split described in Section 2.3.1. We used the PSNR and SSIM metrics to evaluate the model performances. The results depicted in the following paragraph are the average value measured.

The Table 5 shows the results measured for both condition. $MISR_{stat}$ refers to the models trained using the static scenes, $MISR_{dyn}$ refers to the models trained using the dynamic scenes. On average $MISR_{stat}$ models reached a better PSNR on the static scenes, while the performances on the SSIM metrics are equal. When applied on the dynamic test set, both $MISR_{stat}$ and $MISR_{dyn}$ performances decrease and are almost equivalent. Overall, using static scenes for training allows achieving better results in this ideal condition.

Table 5
Average PSNR and SSIM scores (higher scores are better).

Subset	$MISR_{stat}$		$MISR_{dyn}$	
	PSNR	SSIM	PSNR	SSIM
Static	28.62	0.85	28.44	0.85
Dynamic	28.00	0.86	28.01	0.85

Table 6
Average PSNR and SSIM scores (higher scores are better).

Subset	$MISR_{stat}$		$MISR_{dyn}$	
	PSNR	SSIM	PSNR	SSIM
Test	19.17	0.74	19.76	0.76

As in Section 6.1, we also tested the models applied on acquisitions from a different period but over the same regions. We used the test set without discriminating between static and dynamic scenes. The results are shown Table 6. We notice that the average performances dropped drastically across all models. However, the models trained using the dynamic scenes achieved slightly better performances than those trained on the static scenes. Models trained with dynamics scenes were exposed to more variability during the training step, which, we suspect, helps the models be better at generalization. We see that for both training sets, the performances decreased compared with models trained on the whole dataset (cf. Table 2). It can be explained by the fewer data used for training, divided by two in both cases.

6.5. Qualitative results

Fig. 12 shows a few examples of the visual quality of our MISR approach and Fig. 13 shows the effect of the consistency loss in HighRes-net on a few test-set instances. Firstly, Fig. 12 shows a Sentinel-2 (low-res) image, a super resolved image using HighRes-net, and the PlanetScope (high-res) image. Particularly, the first row shows an urban area where the super-resolved image (middle) is significantly sharper than the low-res Sentinel-2 image (left); in this image, it is clear that counting buildings should be easier in the former than in the later. Fig. 13 shows examples of SR with color consistency (second column); we can see that the proposed model enhances the spatial resolution of Sentinel-2 while preserving the spectral content (color appearance). The output on the third column shows the output after the color matching step, which, as we see, it is more similar to the PlanetScope ground truth.

7. Discussion and conclusions

In this work, we proposed the first MISR model for multi-spectral imagery adapting the HighRes-net model of Deudon et al. (2020a). We demonstrated this model over a new compiled dataset of high-res PlanetScope images (4.77 m) and Sentinel-2 lower-res (10 m) time series over 45 different locations using the SpaceNet-7 dataset as a base. Using this data, we show that the HighRes-net model produce better super-resolution images than SISR or bicubic upsampling in terms of the PSNR and SSIM metrics. These results are aligned with the outputs of the Proba-V contest organized by the European Space Agency (ESA) that highlighted the potential of deep learning based super-resolution for Earth observation.

Additionally, we proposed a modification of the HighRes-net architecture and training procedure to deal with the domain shift between PlanetScope and Sentinel-2 images. Taking into account the spectral differences between instruments has been overlooked by previous super-resolution works (e.g. in the SISR work of Salgueiro Romero et al. (2020), authors ignore this issue when working with WorldView and Sentinel-2 images). Nevertheless, the radiometric calibration of Sentinel-2 images is higher (12-bit depth) than the PlanetScope data

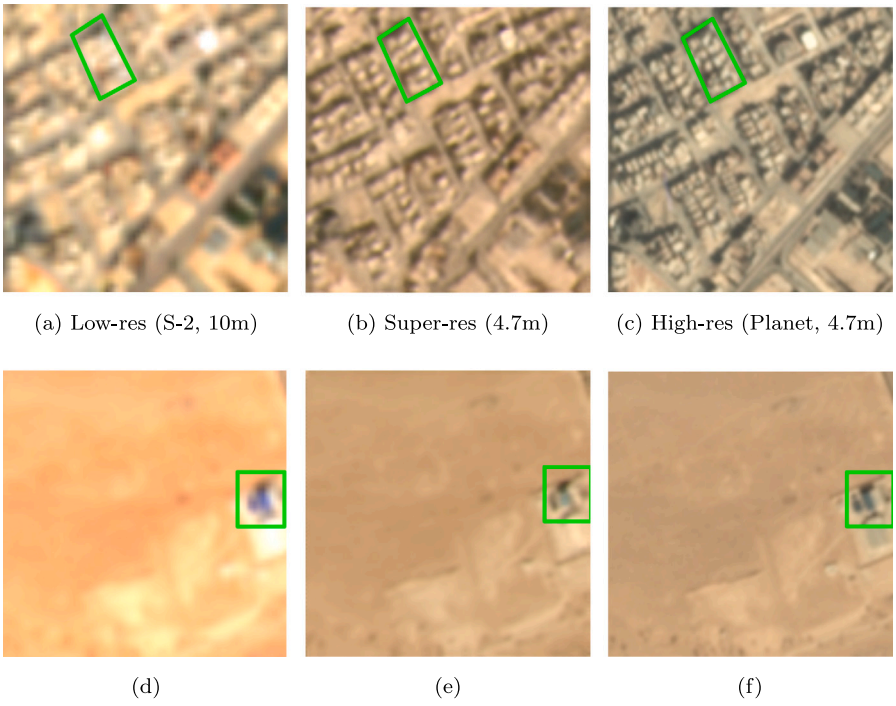


Fig. 12. Patch from a validation-set scene. How many buildings lie within the green polygon? Being more than just a pretty picture, the super-resolved output of HighRes-net 12(b) also better delineates the buildings in urban scenes, hence enabling downstream tasks like building segmentation, with improved accuracy compared to prediction on a single S2 image. This is evidenced qualitatively by the fact that the manual count of buildings in the green polygon in 12(b) is easier to perform than in 12(a). Note that in both examples/rows, the spectra of the super-res output is similar to the high-res PlanetScope reference — an undesirable side-effect if the spectral information of the source low-res instrument is better than that of the high-res instrument. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(8-bit) and Sentinel-2 atmospheric correction is also more accurate due to the dedicated atmospheric correction bands (B1, B9 and B10). Hence, surface reflectance BOA values for Sentinel-2 images are more accurate than in PlanetScope. With our proposed solution we seek to have the best of both worlds: high spatial resolution and good radiometrically calibrated data.

Another important contribution of this work is the demonstration of MISR in a downstream task. In particular, we showed that MISR-fused revisits produce training images that yield better performance of building segmentation models, compared to SISR and bicubic upsampling, in terms of IoU, Recall, Precision, and F1 scores. Interestingly, SISR performs worse than bicubic upsampling, possibly due to artifacts caused by the deep SISR image generator. We attempted a more naive approach of fusion, by concatenating multiple revisits on the channels dimension, and forward-passing them to a Fully Convolutional Net[4] for segmentation. We found that even with this rather simplistic approach, multiple revisits still assist the segmentation task (although we are using the 4 best revisits while HighRes-net receives all the inputs and have to learn to discard the cloudy acquisitions). It is with this context, that we note that MISR is not useful for all downstream tasks. Where are changes in the imagery in a short period of time, far shorter revisits would be required an perhaps SISR would be more suitable.

7.1. Future work

To attempt a fully multi-spectral MISR approach, on all bands of Sentinel-2, ideally a higher resolution reference with the same bands would be needed, at least in a supervised learning setting. Further research into unsupervised MISR is needed to unlock the super-resolution potential of any revisit archive, without depending on near co-located and co-temporal higher resolution imagery.

Although the color consistency loss shows promising results in the preservation of Sentinel-2 spectra, we conclude that a more flexible approach is needed for fusing revisits from dynamic scenes, by attending and fusing features that are static with respect to a certain *anchor* revisit, possibly chosen by the user. This anchor revisit can be any one within the set of available revisits, and the *anchored fusion* model should enhance only the parts of the image that are static with respect to the anchor image, and ignore any dynamic features (e.g. due to weather, vegetation, or urban development).

CRediT authorship contribution statement

Muhammed T. Razzak: Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing. **Gonzalo Mateo-García:** Methodology, Data curation, Writing – original draft, Writing – review & editing. **Gurvan Lecuyer:** Data curation, Writing – review & editing. **Luis Gómez-Chova:** Writing – review & editing. **Yarin Gal:** Conceptualization, Writing – review & editing. **Freddie Kalaitzis:** Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing, Project supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[4] A CNN with no pooling on the spatial dimensions.

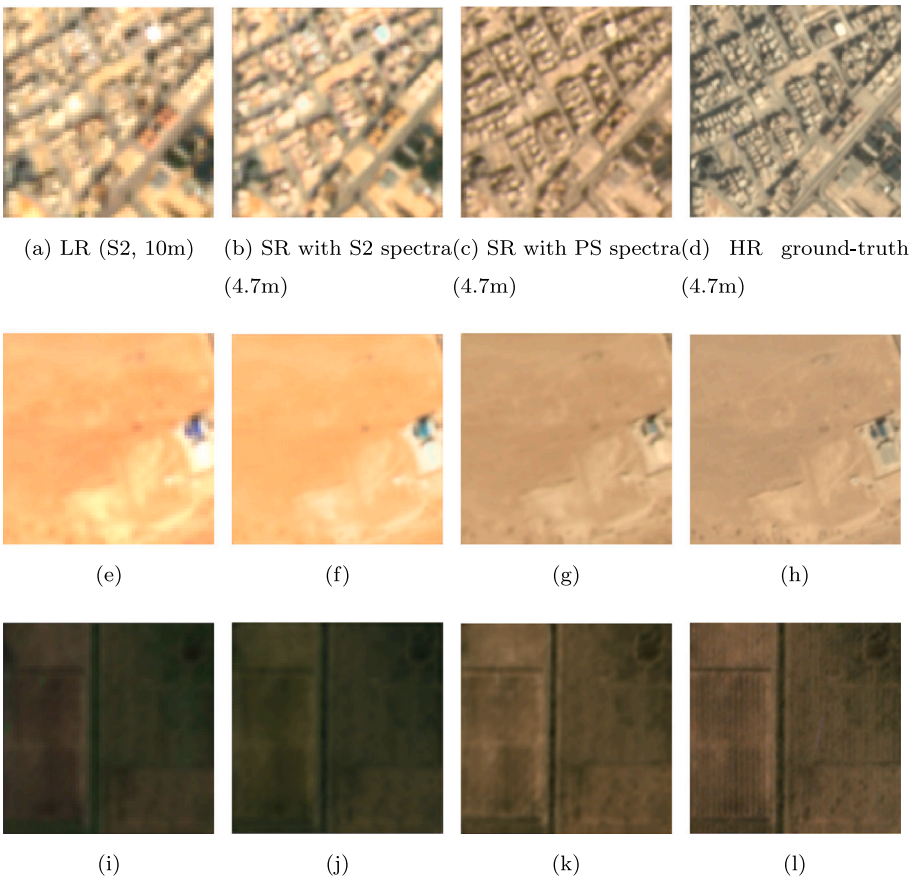


Fig. 13. Example of HighRes-net outputs with the consistency loss, to preserve the spectra of Sentinel-2. 13(a)—13(d): buildings are best delineated in 13(c) but the spectra are no longer consistent with Sentinel-2. 13(b) presents a trade-off that still delineates buildings better than 13(a) while being consistent to S-2 spectra. 13(e)—13(h): example patch from Fig. 12, now with S-2 spectra preserved in 13(f). 13(i)—13(l): example of a failure in an agricultural scene, where HighRes-net with the consistency loss 13(j) has failed to capture the lack of vegetation originally shown in the left side of 13(i). We suspect this is because the high-res ground-truth contains mixed vegetation. In other words, this instance highlights the compromise between the spectra of S-2 and content of PlanetScope. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Data availability

The co-registered dataset produced in this study is publicly available in https://spaceml.org/repo/project/61c0a1b9ff8868000dfb79e1. The code to reproduce our experiments can be found at https://github.com/ESA-PhiLab/RobustMFSRforEO. The PlanetScope images used in this work are publicly available in https://spacenet.ai/sn7-challenge/. Sentinel-2 images were obtained from the Copernicus Open Access Hub.

Acknowledgments

This work has been enabled by Frontier Development Lab (FDL) Europe, a public partnership between the European Space Agency (ESA) at Phi-Lab (ESRIN), Trillium Technologies and the University of Oxford; the project has been also supported by Google Cloud. G.M.G. and L.G.C. are funded by the Spanish Ministry of Science and Innovation, Spain (project PID2019-109026RB-I00). The authors would like to thank the support of James Parr and Jodie Hughes from the Trillium team and to Nicolas Longépé from ESA PhiLab for discussions and comments throughout the development of this work.

Appendix A. Dataset area of interest breakdown

See Table A.1.

Appendix B. HighRes-net architecture

See Table B.1.

Appendix C. Hyper-parameters

C.1. Hyper-parameters of SRResNet

We list a number of the other hyper-parameters used in training. We did not perform a comprehensive hyper-parameter sweep.

1. Optimizer: Adam
2. Learning Rate: 0.0007
3. Learning Rate Decay: The learning rate was reduced when the loss on the validation set plateaued for 2 epochs.
4. Epochs: 50

C.2. Hyper-parameters of HRNet

We list here a number of the hyper-parameters used for training the HRNet (Yuan et al., 2020). We did not perform a comprehensive hyper-parameter sweep.

1. Objective Function: Binary Cross Entropy
2. Optimizer: Adam
3. Learning Rate: 0.0007
4. Learning Rate Decay: The learning rate was reduced when the loss on the validation set plateaued for 2 epochs.
5. Epochs: 50

Table A.1
The subset (45) of SpaceNet-7 AOIs that we used in this work, acquired between December 2019 and January 2020. The high-level breakdown of the types of terrain contained in each scene shows the overall geodiversity of the dataset. Left to right: *% clouds* is the average cloud coverage (SCL = 9) across all revisits; *desert, agri(culture), urban, veg(etation), bare (soils)* indicate the type of terrain; a *usable* revisit is at least %50 cloud-free.

id	Scene	%clouds	desert	agri	urban	veg	bare	revisits		
								usable	total	%
1	0358E-1220N_1433_3310	70		1	1	1		8	13	62
2	1389E-1284N_5557_3054	69	1		1		1	8	13	62
3	0361E-1300N_1446_2989	66		1			1	8	13	62
4	1848E-0793N_7394_5018	66		1	1		1	7	13	54
5	0357E-1223N_1429_3296	65		1	1	1		6	13	46
6	1716E-1211N_6864_3345	62			1			5	13	38
7	1025E-1366N_4102_2726	55		1	1	1		5	13	38
8	1672E-1207N_6691_3363	53			1		1	5	13	38
9	1298E-1322N_5193_2903	56		1	1		1	4	13	31
10	1014E-1375N_4056_2688	49		1	1	1		4	13	31
11	1703E-1219N_6813_3313	58		1	1		1	3	13	23
12	1617E-1207N_6468_3360	24			1	1		3	13	23
13	1439E-1134N_5759_3655	61		1	1	1		5	10	50
14	0566E-1185N_2265_3451	96		1		1		6	7	86
15	0586E-1127N_2345_3680	83		1	1	1		6	7	86
16	1481E-1119N_5927_3715	81		1	1	1		6	7	86
17	0571E-1075N_2287_3888	81			1	1		6	7	86
18	1200E-0847N_4802_4803	80		1	1	1		6	7	86
19	1210E-1025N_4840_4088	95		1	1	1		5	7	71
20	1335E-1166N_5342_3524	84	1		1		1	5	7	71
21	1204E-1204N_4819_3372	74	1		1		1	5	7	71
22	0632E-0892N_2528_4620	67	1		1		1	5	7	71
23	1479E-1101N_5916_3785	67		1		1		5	7	71
24	0434E-1218N_1736_3318	84	1					4	7	57
25	1138E-1216N_4553_3325	71			1		1	4	7	57
26	0331E-1257N_1327_3160	68		1		1		4	7	57
27	1049E-1370N_4196_2710	59			1	1		4	7	57
28	1185E-0935N_4742_4450	33		1	1	1		2	7	29
29	0614E-0946N_2459_4406	29			1	1		2	7	29
30	1209E-1113N_4838_3737	100	1	1	1		1	6	6	100
31	0977E-1187N_3911_3441	100	1		1		1	6	6	100
32	1289E-1169N_5156_3514	99			1		1	6	6	100
33	0368E-1245N_1474_3210	67			1		1	6	6	100
34	1015E-1062N_4061_3941	100			1	1		5	6	83
35	1438E-1134N_5753_3655	92			1	1		5	6	83
36	1276E-1107N_5105_3761	91		1	1		1	5	6	83
37	1296E-1198N_5184_3399	87	1		1		1	5	6	83
38	0924E-1108N_3699_3757	67			1			4	6	67
39	0487E-1246N_1950_3207	98		1		1		3	6	50
40	1538E-1163N_6154_3539	63			1	1		3	6	50
41	1748E-1247N_6993_3202	57			1		1	3	6	50
42	1172E-1306N_4688_2967	56		1	1		1	3	6	50
43	1709E-1112N_6838_3742	44		1	1	1		3	6	50
44	0683E-1006N_2732_4164	58			1	1		2	5	40
45	0760E-0887N_3041_4643	33			1	1		2	5	40

Table B.1
HighRes-net architecture: The ENCODE module converts each input low-res revisit into an encoding, and it inputs 6 channels, that is, 3 channels (RGB) per input image (low-res revisit + reference frame concatenated in the channels dimension). The FUSE module is an operator that is applied recursively of a pair of encodings, until one encoding remains. An encoding can be either the output of the ENCODE or the FUSE module. It inputs 64 channels per input encoding, that is, 128 channels for a pair of encodings concatenated on the channels dimension. The DECODE module is a learned upsampling operator (contrary to the non-learned bilinear or bicubic upsampling), through a transpose convolution layer that outputs 3 channels (RGB). Note that the ConvTranspose2d stride also decides the upsampling factor, which must be an integer, hence an optional Upsample layer can further upscale the super-resolved image by a fractional factor, if needed.

Module	Layers	Parameters
ENCODE	`Conv2d(in=6, out=64, k=3, s=1, p=1)`	1 216
	`PReLU`	1
	`ResidualBlock(64)`	73,858
	`ResidualBlock(64)`	73,858
	`Conv2d(in=64, out=64, k=3, s=1, p=1)`	36,928
FUSE	`ResidualBlock(128)`	295,170
	`Conv2d(in=128, out=64, k=3, s=1, p=1)`	73,792
	`PReLU`	1
DECODE	`ConvTranspose2d(in=64, out=64, k=3, s=1)`	36,928
	`PreLU`	1
	`Conv2d(in=64, out=3, k=1, s=1)`	65
RESIDUAL (OPTIONAL)	`Upsample(scale_factor=3.0, mode='bicubic')`	0
		591,818 (TOTAL)

References

Anwar, S., Khan, S., Barnes, N., 2020. A deep journey into super-resolution: A survey. arXiv:1904.07523 [Cs].

Bahat, Y., Michaeli, T., 2020. Explorable super resolution. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 2716–2725.

Bhadra, S., Kelkar, V.A., Brooks, F.J., Anastasio, M.A., 2020. On hallucinations in tomographic image reconstruction. arXiv:2012.00646.

Bhat, G., Danelljan, M., Van Gool, L., Timofte, R., 2021. Deep burst super-resolution. arXiv preprint arXiv:2101.10997.

Boyd, S., Parikh, N., Chu, E., 2011. Distributed Optimization and Statistical Learning Via the Alternating Direction Method of Multipliers. Now Publishers Inc.

DeTone, D., Malisiewicz, T., Rabinovich, A., 2016. Deep image homography estimation. arXiv preprint arXiv:1606.03798.

Deudon, M., Kalaitzis, A., Goytom, I., Arefin, M.R., Lin, Z., Sankaran, K., Michalski, V., Kahou, S.E., Cornebise, J., Bengio, Y., 2020a. HighRes-net: Recursive fusion for multi-frame super-resolution of satellite imagery. arXiv:2002.06460 [Cs, Eess, Stat].

Deudon, M., Kalaitzis, A., Goytom, I., Arefin, M.R., Lin, Z., Sankaran, K., Michalski, V., Kahou, S.E., Cornebise, J., Bengio, Y., 2020b. HighRes-net: Recursive fusion for multi-frame super-resolution of satellite imagery. arXiv:2002.06460 [Cs, Eess, Stat].

Dong, C., Loy, C.C., He, K., Tang, X., 2015. Image super-resolution using deep convolutional networks. arXiv:1501.00092 [Cs], version : 3.

Dong, C., Loy, C.C., Tang, X., 2016. Accelerating the super-resolution convolutional neural network. arXiv:1608.00367 [Cs] version: 1.

Jiang, K., Wang, Z., Yi, P., Wang, G., Lu, T., Jiang, J., 2019. Edge-enhanced GAN for remote sensing image superresolution. IEEE Trans. Geosci. Remote Sens. 57 (8), 5799–5812. http://dx.doi.org/10.1109/TGRS.2019.2902431.

Kawulok, M., Tarasiewicz, T., Nalepa, J., Tyrna, D., Kostrzewa, D., 2021. Deep learning for multiple-image super-resolution of sentinel-2 data. In: 2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS. pp. 3885–3888. http://dx.doi.org/10.1109/IGARSS47720.2021.9553243.

Latte, N., Lejeune, P., 2020. PlanetScope radiometric normalization and sentinel-2 super-resolution (2.5 m): A straightforward spectral-spatial fusion of multi-satellite multi-sensor images using residual convolutional neural networks. Remote Sens. 12 (15), http://dx.doi.org/10.3390/rs12152366, URL https://www.mdpi.com/2072-4292/12/15/2366.

Ledig, C., Theis, L., Huszár, F., Caballero, J., Cunningham, A., Acosta, A., Aitken, A., Tejani, A., Totz, J., Wang, Z., Shi, W., 2017. Photo-realistic single image super-resolution using a generative adversarial network. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 105–114. http://dx.doi.org/10.1109/CVPR.2017.19.

Li, J., Cui, R., Li, B., Song, R., Li, Y., Dai, Y., Du, Q., 2020. Hyperspectral image super-resolution by band attention through adversarial learning. IEEE Trans. Geosci. Remote Sens. 1–15. http://dx.doi.org/10.1109/TGRS.2019.2962713.

Lugmayr, A., Danelljan, M., Gool, L.V., Timofte, R., 2020. SRFlow: Learning the super-resolution space with normalizing flow. arXiv:2006.14200.

Märtens, M., Izzo, D., Krzic, A., Cox, D., 2019. Super-resolution of PROBA-V images using convolutional neural networks. Astrodynamics 3 (4), 387–402.

Mateo-García, G., Laparra, V., López-Puigdollers, D., Gómez-Chova, L., 2020. Cross-sensor adversarial domain adaptation of landsat-8 and proba-V images for cloud detection. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 1. http://dx.doi.org/10.1109/JSTARS.2020.3031741.

Molini, A.B., Valsesia, D., Fracastoro, G., Magli, E., 2020. DeepSUM: Deep neural network for super-resolution of unregistered multitemporal images. IEEE Trans. Geosci. Remote Sens. 58 (5), 3644–3656. http://dx.doi.org/10.1109/TGRS.2019.2959248.

Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., Dormann, C., Cornu, G., Viennois, G., Bayol, N., Lyapustin, A., Gourlet-Fleury, S., Pélissier, R., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. Nature Commun. 11 (1), 4540. http://dx.doi.org/10.1038/s41467-020-18321-y.

Ruescas, A.B., Hieronymi, M., Mateo-Garcia, G., Koponen, S., Kallio, K., Camps-Valls, G., 2018. Machine learning regression approaches for colored dissolved organic matter (CDOM) retrieval with S2-MSI and S3-OLCI simulated data. Remote Sens. 10 (5), 786. http://dx.doi.org/10.3390/rs10050786.

Salgueiro Romero, L., Marcello, J., Vilaplana, V., 2020. Super-resolution of sentinel-2 imagery using generative adversarial networks. Remote Sens. 12 (15), 2424. http://dx.doi.org/10.3390/rs12152424.

Salvetti, F., Mazzia, V., Khaliq, A., Chiaberge, M., 2020. Multi-image super resolution of remotely sensed images using residual feature attention deep neural networks. arXiv preprint arXiv:2007.03107.

Sauzède, R., Johnson, J.E., Claustre, H., Camps-Valls, G., Ruescas, A.B., 2020. Estimation of oceanic particulate organic carbon with machine learning. ISPRS Ann. Photogramm. Remote Sens. Spat. Inf. Sci. 5 (2), 949–956. http://dx.doi.org/10.5194/isprs-annals-V-2-2020-949-2020, ADS Bibcode: 2020ISPAn.5.2..949S.

Shermeyer, J., Van Etten, A., 2019. The effects of super-resolution on object detection performance in satellite imagery. In: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). IEEE, Long Beach, CA, USA, pp. 1432–1441. http://dx.doi.org/10.1109/CVPRW.2019.00184.

Svendsen, D.H., Martino, L., Campos-Taberner, M., García-Haro, F.J., Camps-Valls, G., 2018. Joint Gaussian processes for biophysical parameter retrieval. IEEE Trans. Geosci. Remote Sens. 56 (3), 1718–1727. http://dx.doi.org/10.1109/TGRS.2017.2767205.

Tasar, O., Happy, S.L., Tarabalka, Y., Alliez, P., 2020. ColorMapGAN: Unsupervised domain adaptation for semantic segmentation using color mapping generative adversarial networks. IEEE Trans. Geosci. Remote Sens. 1–16. http://dx.doi.org/10.1109/TGRS.2020.2980417.

Van Etten, A., Hogan, D., Martinez-Manso, J., Shermeyer, J., Weir, N., Lewis, R., 2021. The multi-temporal urban development SpaceNet dataset. arXiv:2102.04420 [Cs].

Wang, Z., Bovik, A., Sheikh, H., Simoncelli, E., 2004. Image quality assessment: from error visibility to structural similarity. IEEE Trans. Image Process. 13 (4), 600–612. http://dx.doi.org/10.1109/TIP.2003.819861, Conference Name: IEEE Transactions on Image Processing.

Wang, Z., Chen, J., Hoi, S.C., 2020. Deep learning for image super-resolution: A survey. IEEE Trans. Pattern Anal. Mach. Intell..

Wang, J., Sun, K., Cheng, T., Jiang, B., Deng, C., Zhao, Y., Liu, D., Mu, Y., Tan, M., Wang, X., Liu, W., Xiao, B., 2019. Deep high-resolution representation learning for visual recognition. TPAMI.

Wang, X., Yu, K., Wu, S., Gu, J., Liu, Y., Dong, C., Loy, C.C., Qiao, Y., Tang, X., 2018. ESRGAN: Enhanced super-resolution generative adversarial networks. arXiv:1809.00219.

Wolanin, A., Camps-Valls, G., Gómez-Chova, L., Mateo-García, G., van der Tol, C., Zhang, Y., Guanter, L., 2019. Estimating crop primary productivity with Sentinel-2 and Landsat 8 using machine learning methods trained with radiative transfer simulations. Remote Sens. Environ. 225, 441–457. http://dx.doi.org/10.1016/j.rse.2019.03.002.

Wronski, B., Garcia-Dorado, I., Ernst, M., Kelly, D., Krainin, M., Liang, C.-K., Levoy, M., Milanfar, P., 2019. Handheld multi-frame super-resolution. arXiv preprint arXiv:1905.03277.

Yuan, Y., Chen, X., Wang, J., 2020. Object-contextual representations for semantic segmentation. In: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part VI 16. Springer, pp. 173–190.

Zhang, L., Chen, D., Ma, J., Zhang, J., 2020. Remote-sensing image superresolution based on visual saliency analysis and unequal reconstruction networks. IEEE Trans. Geosci. Remote Sens. 1–17. http://dx.doi.org/10.1109/TGRS.2019.2960781.